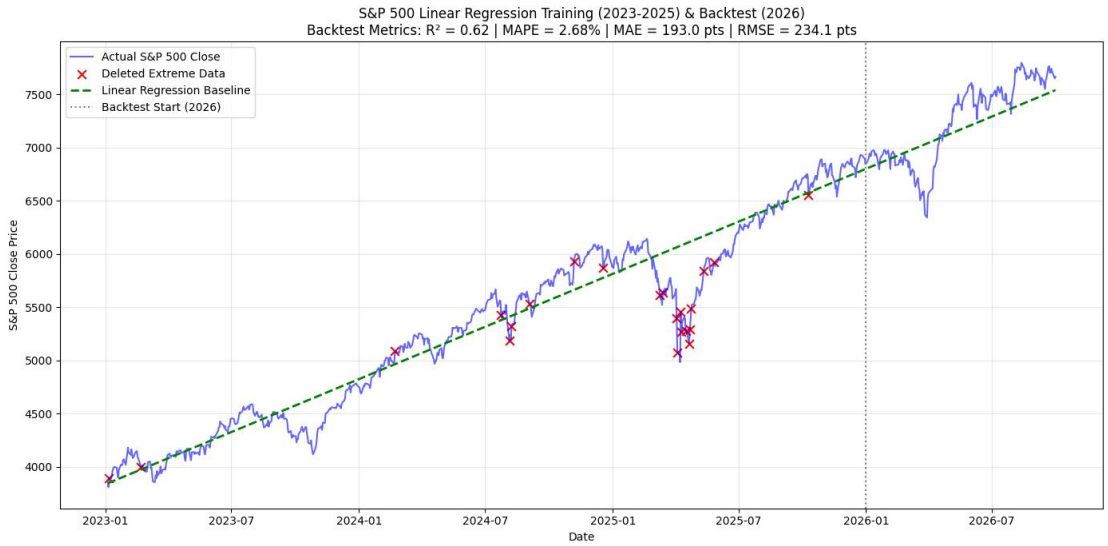
Appendix – Linear Regression Model

This appendix illustrates the overview of data analysis and modelling being applied by Linear Regression model in this study



Fetching S&P 500 data...

[**********100%**********] 1 of 1 completed

--- Extreme Data Removed (Daily Return > 2.0%) ---

--- No. of Extreme Data Removed : 22---

Price	Close	Daily_Return
Ticker	^GSPC	
Date		
2023-01-06	3895.080078	2.284078
2023-02-21	3997.340088	-2.004123
2024-02-22	5087.029785	2.112288
2024-07-24	5427.129883	-2.314909
2024-08-05	5186.330078	-2.996880
2024-08-08	5319.310059	2.304261
2024-09-03	5528.930176	-2.115107
2024-11-06	5929.040039	2.529593
2024-12-18	5872.160156	-2.949285
2025-03-10	5614.560059	-2.697309

2025-03-14	5638.939941	2.126587
2025-04-03	5396.520020	-4.839563
2025-04-04	5074.080078	-5.974961
2025-04-09	5456.899902	9.515388
2025-04-10	5268.049805	-3.460758
2025-04-16	5275.700195	-2.240837
2025-04-21	5158.200195	-2.356749
2025-04-22	5287.759766	2.511720
2025-04-24	5484.770020	2.025911
2025-05-12	5844.189941	3.255878
2025-05-27	5921.540039	2.045906
2025-10-10	6552.509766	-2.711167

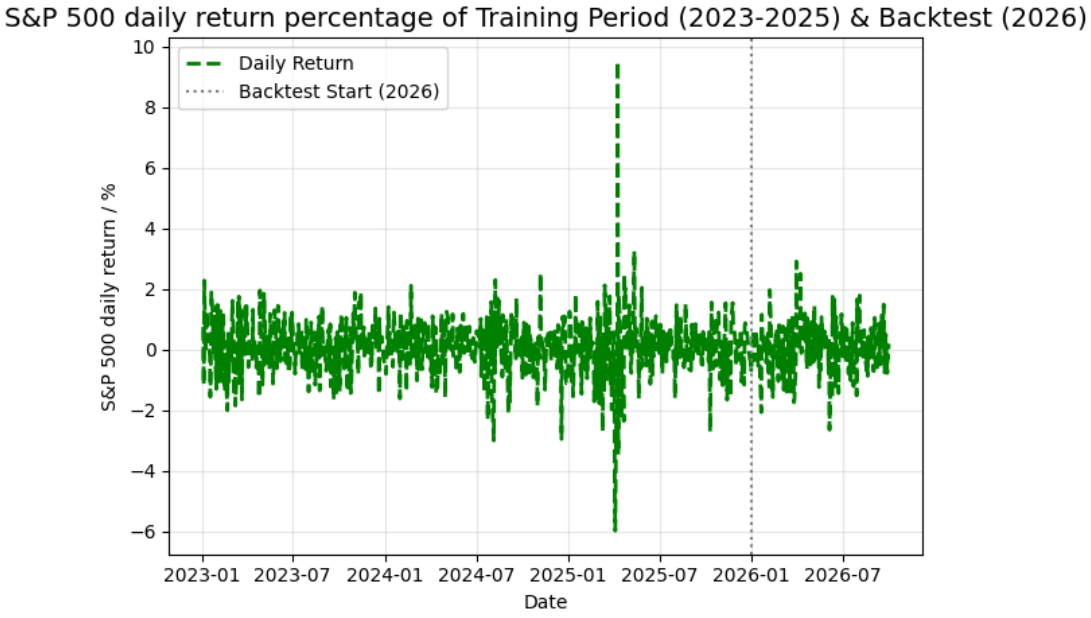
--- Backtest (2026) Evaluation Metrics ---

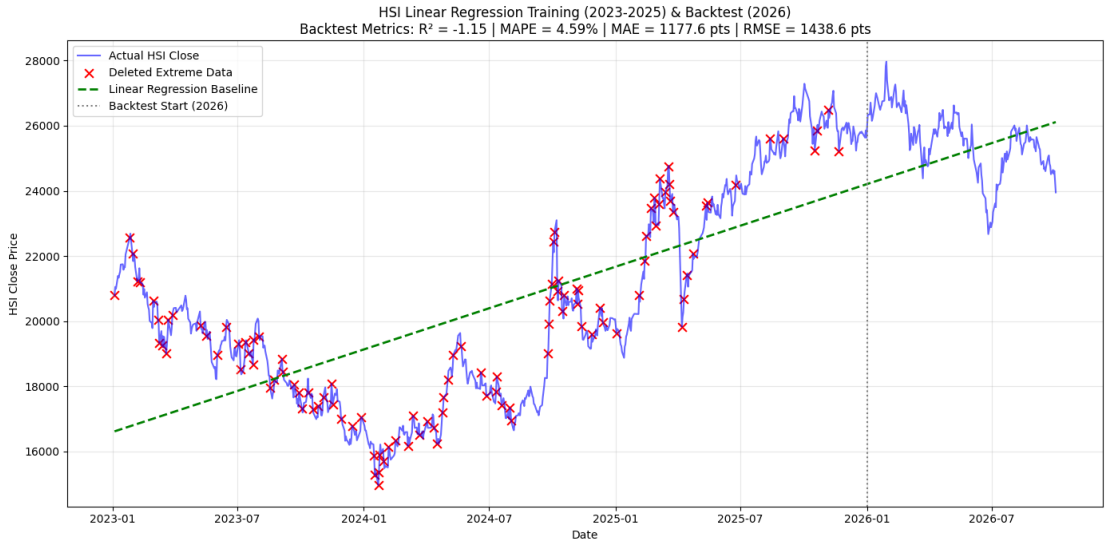
R2 Score : 0.6190

MAPE : 2.68%

MAE : 193.00 points

RMSE : 234.14 points





Fetching HSI data...

--- Extreme Data Removed (Daily Return > 2%) ---

--- No. of Extreme Data Removed : 109---

Price	Close	Daily_Return
Ticker	^HSI	
Date		
2023-01-04	20793.109375	3.215741
2023-01-26	22566.779297	2.368506
2023-01-30	22069.730469	-2.728955
2023-02-06	21222.160156	-2.023550
2023-02-10	21190.419922	-2.006716
...	...	...
2025-09-01	25617.419922	2.152520
2025-10-17	25247.099609	-2.477586
2025-10-20	25858.830078	2.422973
2025-11-06	26485.900391	2.122543
2025-11-21	25220.019531	-2.382571

[109 rows x 2 columns]

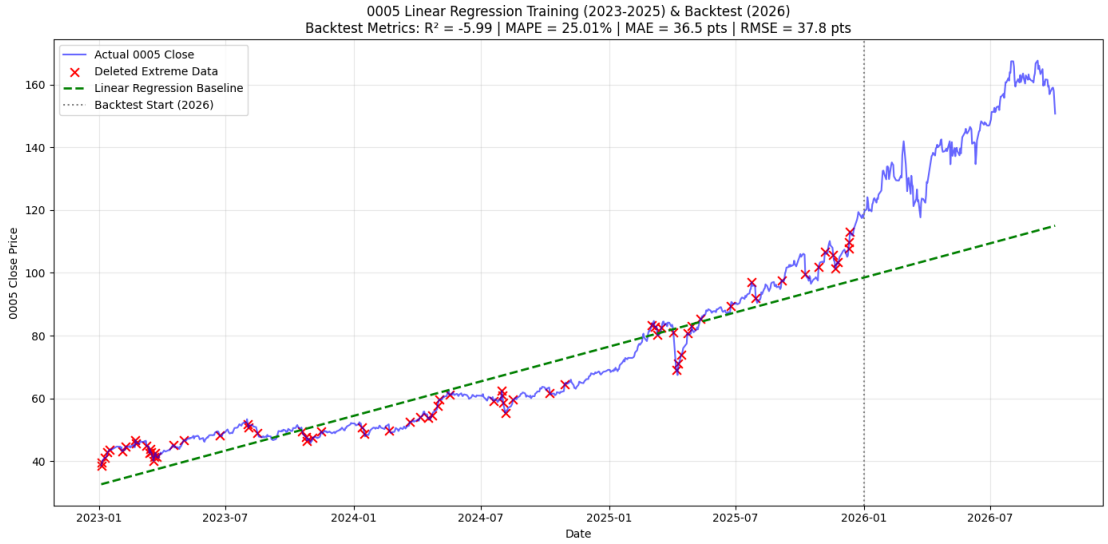
--- Backtest (2026) Evaluation Metrics ---

R2 Score : -1.1465

MAPE : 4.59%

MAE : 1177.65 points

RMSE : 1438.64 points



Fetching 0005 data...

--- Extreme Data Removed (Daily Return > 2%) ---

--- No. of Extreme Data Removed : 70---

Price	Close	Daily_Return
Ticker	0005.HK	
Date		
2023-01-04	38.540058	2.669392
2023-01-05	39.503563	2.500009
2023-01-09	41.083706	2.107280
2023-01-13	42.779472	2.587807
2023-01-16	43.665894	2.072071
...	...	...
2025-11-21	101.253876	-2.323416
2025-11-24	103.373375	2.093252
2025-12-10	107.612350	2.009133
2025-12-11	109.828171	2.059076
2025-12-12	113.007416	2.894745

[70 rows x 2 columns]

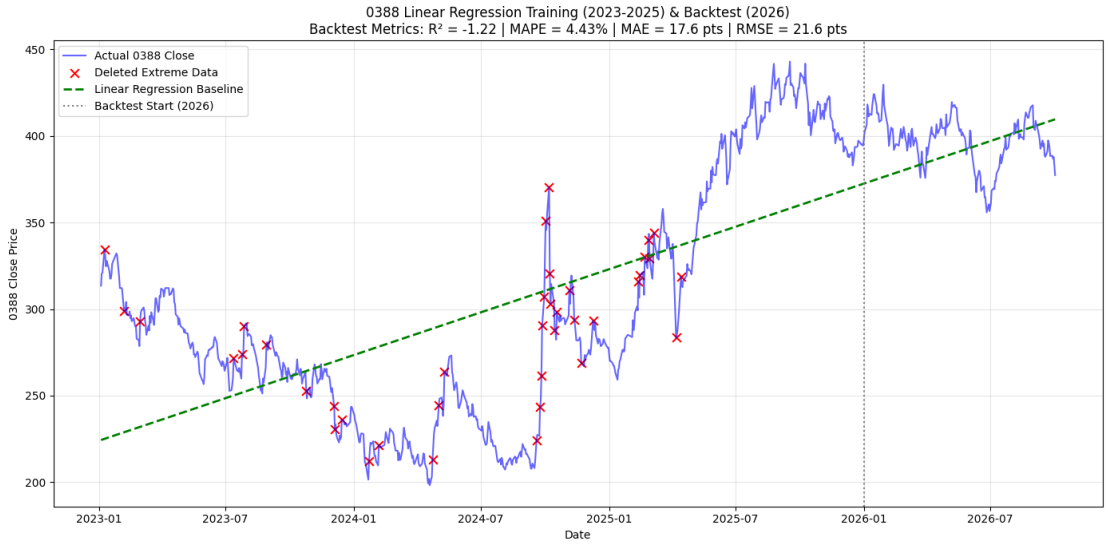
--- Backtest (2026) Evaluation Metrics ---

R2 Score : -5.9855

MAPE : 25.01%

MAE : 36.49 points

RMSE : 37.83 points



Fetching 0388 data...

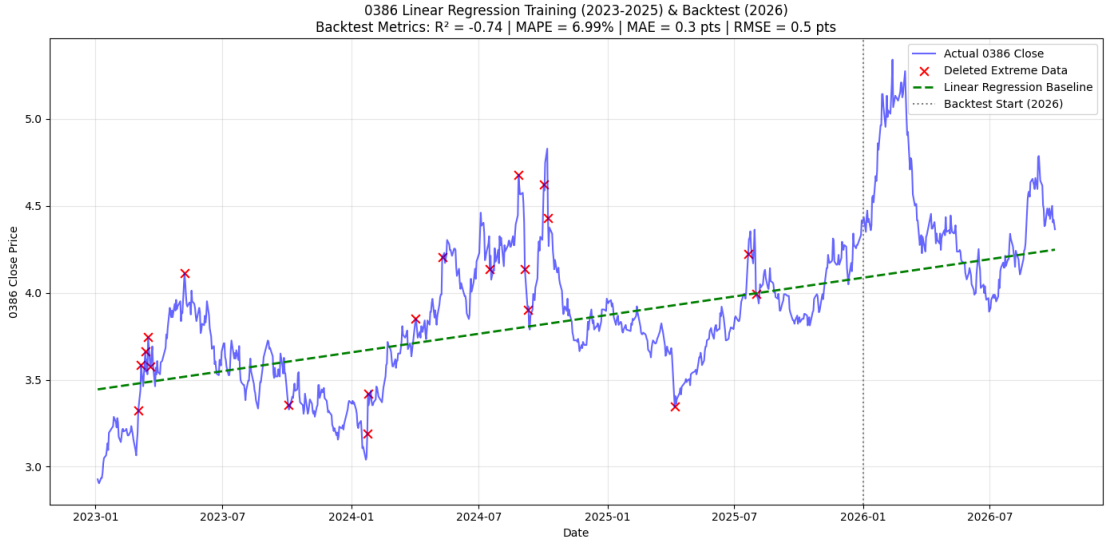
--- Extreme Data Removed (Daily Return > 4%) ---

--- No. of Extreme Data Removed : 39---

Price	Close	Daily_Return
Ticker	0388.HK	
Date		
2023-01-09	334.248901	4.139086
2023-02-06	298.999542	-4.199779
2023-03-01	292.976990	5.149370
2023-07-13	271.502014	4.335849
2023-07-25	274.009338	5.444508
2023-07-28	290.306641	5.259752
2023-08-29	279.494873	5.022185
2023-10-25	252.945602	-4.068971
2023-12-04	244.035233	-4.074343
2023-12-05	230.578751	-5.514155
2023-12-15	236.215897	4.421211
2024-01-23	212.212433	5.324902
2024-02-06	221.486511	5.637466
2024-04-23	213.185349	4.909088
2024-05-02	244.590469	5.246411

2024-05-10	263.987762	7.605416
2024-09-19	224.147644	5.772646
2024-09-24	243.532349	7.385886
2024-09-26	261.599670	5.864430
2024-09-27	290.394470	11.007200
2024-09-30	306.956177	5.703176
2024-10-02	350.807037	14.285707
2024-10-07	370.568146	4.678345
2024-10-08	320.694855	-13.458602
2024-10-09	303.003967	-5.516424
2024-10-15	287.759644	-5.207696
2024-10-18	298.298920	5.666677
2024-11-05	310.908386	4.955536
2024-11-12	293.970276	-4.171775
2024-11-22	268.939514	-4.222525
2024-12-09	293.217468	5.770524
2025-02-12	315.989838	5.597495
2025-02-14	319.753815	4.297101
2025-02-21	330.104889	7.081793
2025-02-26	339.891418	5.061101
2025-02-28	329.352112	-4.109587
2025-03-06	344.031799	5.178367
2025-04-07	283.447052	-14.260973
2025-04-14	318.758636	6.914219

--- Backtest (2026) Evaluation Metrics ---
R2 Score : -1.2222
MAPE : 4.43%
MAE : 17.65 points
RMSE : 21.59 points



--- Extreme Data Removed (Daily Return > 4%) ---

--- No. of Extreme Data Removed : 20---

Price	Close	Daily_Return
Ticker	0386.HK	
Date		
2023-03-03	3.324669	4.067000
2023-03-07	3.584528	4.222218
2023-03-13	3.660957	5.739510
2023-03-17	3.745029	6.060599
2023-03-20	3.576885	-4.489785
2023-05-08	4.111889	5.905521
2023-10-03	3.353570	-5.140186
2024-01-24	3.188369	4.324318
2024-01-25	3.419650	7.253891
2024-04-02	3.849170	4.954935
2024-05-10	4.204352	5.383034
2024-07-17	4.135691	-4.382466
2024-08-27	4.678500	5.232559
2024-09-05	4.135368	-6.073875
2024-09-09	3.904142	-5.591424
2024-10-02	4.624496	7.438019
2024-10-08	4.428845	-8.287288
2025-04-07	3.343867	-8.292692
2025-07-21	4.224501	5.529953

2025-08-01 3.993905 -5.869569

--- Backtest (2026) Evaluation Metrics ---
R2 Score : -0.7391
MAPE : 6.99%
MAE : 0.33 points
RMSE : 0.45 points